MITA — Multimodal Interactive Botanical Garden
Short Description of the Project Concept

Course: Multimodal Interactive Technologies and Applications

1 Vision

MITA (Multimodal Interactive Tour Assistant) is a browser-based, immersive botanical garden combining **WebVR panoramic exploration** with a **conversational, voice-driven assistant**. Users walk through 360° panoramic scenes, point at plants in 3D and *talk* to the system in natural language, turning a traditional static-label exhibit into a fluid multimodal dialogue between user, environment and content.

2 Characterizing Features

- **Immersive 3D environment.** Built on **A-Frame (WebVR)** and rendered in-browser. Five panoramic scenes (`scene-1`... `scene-5`) are linked by clickable arrow portals; plant specimens (`Lavandula`, `Nephrolepis`, `Santolina`, `Opuntia`...) are positioned `glTF` models inspectable from any angle.
- **Multimodal input.** Three integrated channels: (i) *gaze / pointer raycasting* to select a plant in 3D; (ii) *speech (STT)* via the Web Speech API for hands-free queries (*"What is this used for?"*, *"Compare it with the lavender."*); (iii) *click / touch fallback* for accessibility and noisy environments.
- **Multimodal output.** Coordinated combination of a *spatial highlight* (glow / outline) on the selected plant, *adaptive 2D HUD cards* (botanical, medicinal, comparison, fallback), and *TTS narration* so the user never needs to leave the VR view to read.
- **LLM-driven dialogue manager.** A Node.js / Express + WebSocket backend routes each user turn through an LLM performing **intent parsing** (`query_info`, `query_attribute`, `compare`) and **slot extraction** (`medicinal`, `botanical`, `toxicity`, `drought`, `feng_shui`), interpreted against the selected plant, dialogue history and the structured `plants.json` database.
- **State-machine interaction.** A finite set of WebSocket message types (`query`, `clarify`, `disambiguate`, `response`, `comparison`, `fallback`, `clarify_retry`, `error`) and a per-client session tracking visited plants, pending disambiguations and history allow the assistant to reason contextually across turns.
- **Robust disambiguation.** When raycasting returns overlapping plants, candidates are ranked by depth, labelled *A / B / C* and announced verbally; the user resolves by voice (*"the one in front"*, *"A"*) or by clicking the overlay, with up to two clarification retries before falling back to pointer selection.

3 Target Users and Why It Is Multimodal

The system targets **casual museum visitors, botany students and accessibility-oriented users** exploring a botanical collection on desktop, tablet or VR headset. Rather than reading long labels, users ask focused questions (*"Can I keep this around my cat?"*, *"How does it compare to the cactus?"*) and receive concise spoken answers paired with a visual card. MITA is multimodal **by design**: every interaction loop fuses at least two input modalities (point + speak) and three output modalities (3D highlight + card + voice), and degrades gracefully — if STT fails the user can click; if the LLM is uncertain a structured A/B overlay takes over; if a slot is missing, a *fallback card* proposes alternatives. This redundancy keeps the experience fluent across users, devices and ambient conditions.